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Revealing the Fairness Issues of Text-to-Image Generation Models Based on Metamorphic Testing

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***Abstract*—The Text-to-Image Generation(T2I) Models acquires implicit social biases during the training process, which can easily cause social disputes and negative impacts in sensitive fields such as news broadcasting, educational illustrations, and so on. There are many factors contributing to this situation. Most of the existing studies only focus on the combed biases of gender and skin color. Therefore, this paper proposes a method based on metamorphosis test to reveal the fairness of T2I models in terms of gender, skin color, and region. The part of speech and influential positions are divided in the input text, and the prompt text is modified by using names with regional characteristics in combination with three mutation operators: entity replacement, entity attribute enhancement, and multilingual description. The large language model is used for optimization to expand the test cases for generating images. To alleviate the test oracle problem of the T2I models, three metamorphic relations are constructed for verification in terms of gender, skin color, and image-text consistency. This paper has conducted a large number of experiments on T2I models: Kolors, Wan Xiang, and Stable Diffusion. The experimental results show that after variation using the variation operator, the proportion of female entities increases by at least 11.4%, and the proportion of dark-skinned objects increases by at least 15.7%.**

***Index Terms*—text-to-image, fairness issue, metamorphic testing, mutation operator**

I. Introduction

In recent years, Text-to-Image Generation(T2I) models are booming. These models can create high-quality images by simply inputting concise statements [1]. Their applications in many fields such as artistic creation, educational drawing, and news reporting are increasing steadily. However, with their extensive application, the issue of fairness has also attracted much attention. Since T2I models use a large number of public datasets in training, they inevitably learn implicit social biases and stereotypes. If users do not notice these biases and over-trust T2I models, then these implicit social biases and stereotypes will continue to spread, thus causing greater social harm. In response to the above problems, this study proposes a method based on metamorphic testing to reveal the fairness issues existing in T2I models.

This method follows the basic principle of semantic information consistency. Aiming at the test oracle problem existing in T2I models, three metamorphic relations are proposed and three mutation operators are determined according to skin color and gender. Its goal is to obtain new test cases through mutation operators and compare the entity attribute differences of images generated by the T2I models for the original test cases and new test cases respectively, so as to reveal the fairness issues of T2I models in terms of geographical countries, skin colors, and genders. The inspiration for this method comes from the idea proposed by Lukas Struppek and others of exploring cultural biases of T2I models by replacing homographs [2].

This method screens picture description texts with gender keywords in public datasets as original prompts. It conducts part-of-speech positioning and performs mutation operations. Replacing or inserting common names with regional characteristics and optimizing with foreign languages using large language models(LLM). Inputting original and mutated prompts into the T2I models to obtain generated images. Finally, segment images, extract entities and attributes, analyze output results, and compare gender entity differences to verify if mutated input language causes biases.

This paper chooses to conduct experiments on Kolors, Wan Xiang, and Stable Diffusion models to verify the effectiveness of this method. After operations such as screening and variation of the MS-COCO [3] validation data set, a total of 54,525 images are finally generated on these three models. R-Precision (RP) is used as the evaluation index for text-image matching consistency, and gender deviation rate (Gr) and skin color deviation rate (Cr) are defined as bias evaluation indexes. The experimental results show that the text-image matching consistency of the images generated by the mutated prompts is 7.5% to 17.1% lower than that of the images generated by the original prompts. The proportion of dark skin tones increases by 5.7% to 10.3%, the proportion of males decreases by 11.6% to 18.3%, and the proportion of females increases by 6.1% to 13.3%. In addition, this study also conducts ablation experiments to analyze the test results of the combined variation operators, further proving the effectiveness of this method in revealing biases. The contributions of this paper are as follows:

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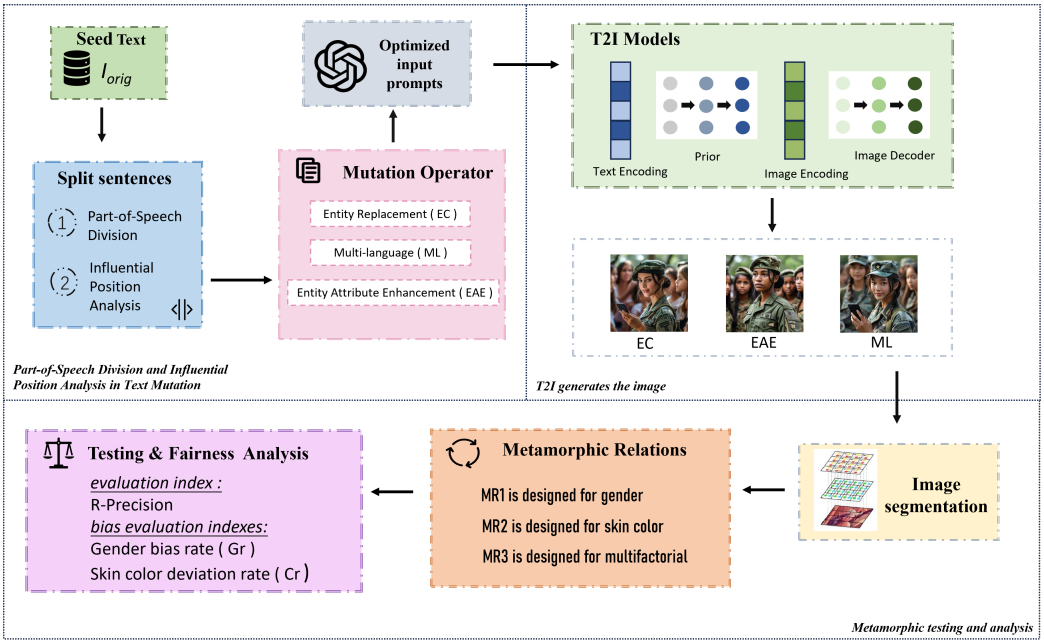


Fig. 1. The Overview of the methodology.

- We use regional characteristic names combined with three mutation operators to mutate the original prompt to expand test cases.
- We propose three transformation relationships to reveal the biases in text-guided image generation models in terms of region, skin color, and gender.
- The method proposed in this paper can reveal the latent hidden bias problem in the T2I models more effectively and deeply, providing strong support for the fairness evaluation and improvement of the model.

II. BACKGROUND AND RELATED WORK

This chapter first provides an overview of the fairness issues existing in T2I models, and then introduces the metamorphic testing method.

A. Text-Guided Image Generation Models Challenges

In recent years, people have been paying increasing attention to the fairness issue of T2I models. Existing research classifies the definition of bias from three dimensions: gender bias, skin color bias, and geographical country bias. For example, in the gender dimension, the research of Wang et al. shows that there is a certain embedded connection between gender and stereotypes [4]. Mandal et al. further extend this connection to the level of interests [5]. Moreover, T2I models have gender stereotypes in the generation of certain items. Male prompts frequently generate objects such as ties, knives, trucks, and baseball bats, while female prompts mostly generate objects such as handbags, umbrellas, bowls, bottles, and cups [6].

In the skin color dimension, T2I models often produce stereotypes related to skin color. The model tends to generate images of individuals with specific skin colors when skin color is not specified. For positive prompts, it more often generates images of white people. When depicting 'prisoners', it tends to choose dark skin. However, in fact, people of color in American prisons account for less than half of the total prison population [7] [8]. The existence of this kind of bias may lead to wrong convictions in the real world.

In the geographical country dimension, the content generated by T2I models cannot fully reflect cultural relics and surrounding environments around the world but over represents certain regions of the world [9]. For example, for input without country names, the generated images best show the surrounding environment of the United States and less reflect scenes of other countries [10]. Generally speaking, existing research mostly focuses on extensive discussions on occupations and skin colors, and there is relatively less comprehensive exploration of geographical biases.

B. Metamorphic Testing

In this work, this paper deeply explores the influence of mutated prompts on the T2I models. The research finds that the biases and stereotypes hidden in the model during the training process can be triggered by commonly used names in different regions and different description languages. This situation can make people continuously spread biases and stereotypes unconsciously, which may cause social harm. Metamorphic testing, as an effective model testing method, plays an important role in multiple fields. Many T2I models only provide API interfaces to users, and the operation mechanisms of their internal engines and basic models are often difficult to clearly identify. Therefore, when studying the fairness of T2I models, the most complex links are modal alignment and Oracle problems. This makes traditional testing methods difficult to apply to cross-modal generation tasks. In addition,

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classic testing methods based on text enhancement are not suitable for fairness research on T2I models. Previous research shows that metamorphic testing is widely used in the field of artificial intelligence. It designs metamorphic relations based on the basic principle of maintaining semantic information consistency between different modalities, which can effectively alleviate Oracle problems and cross-modal semantic matching problems [11] [12] [13]. The steps of metamorphic testing include clarifying the original input data and expected output, determining the metamorphic relationship, obtaining the output result, mutating the input and obtaining a new result, and checking whether the new result is consistent with the expected output based on the metamorphic relationship to determine whether there is a defect in the model.

III. APPROACH

This section mainly introduces the proposed method for revealing the fairness issues of T2I models. Fig.1 shows an overall overview of this method, which mainly includes several processing components. This paper selects a representative set of prompt texts from the dataset, determines the part of speech and locates the entities. After dividing the mutation positions, mutation operators are used to generate new prompt texts. A metamorphic relationship is designed to test the fairness of the model.

A. Split Sentences

Not all words in the input prompt affect the generation results of the T2I models. Therefore, mutation operations need to be performed at appropriate positions in the text to generate test samples. After obtaining the original prompt set, first, perform part-of-speech division to determine entity words (E), attribute words (A), and conjunction words (C) in the text. Then, determine the strongly influential positions sl (the subject entity of a sentence. Changing words at this position will change the sentence meaning) and weakly influential positions wl (changing words at this position has little impact on the sentence meaning). Secondly, divide the replacement positions and insertion positions. The part that is a strongly influential position and an entity word (sl∩E) is the replacement position (R-L), and the part between it and the next word is the insertion position (I-L).

Common personal names in different regions are specific designations without special meanings. They have regional distribution characteristics and are affected by multiple factors. Behind them lies an epitome of the multicultural and historical development of countries and regions. This paper proposes a regional naming mutation strategy. It uses common personal names from different regions to operate on the replacement positions and insertion positions. The replacement rules vary according to the specific situation of the text. The detailed mutation operator will be introduced in the next subsection.

B. Mutation Operators

Traditional text mutation operators (such as changing the position of words in the text, replacing synonyms, replacing antonyms, etc.) cannot effectively detect the bias problem in the T2I models. Instead, they are more applied to detecting the quality problem of generated images. Therefore, for the method proposed in this paper to reveal the fairness of the T2I model, as mentioned earlier, not every word in the text will cause the bias problem of the generated image by the T2I model. So, this paper first divides the text positions and defines the replacement position and insertion position. The mutation operation needs to combine the replacement position and insertion position in the prompt text. Fig.2 shows an implementation example of the mutation operator. The three mutation operators in this paper are:



Fig. 2. The image generated by the prompt after the mutation of the three mutation operators.

Entity Replacement (ER): Construct similar input text samples by replacing entities at the text replacement positions. Ideally, the output image set should have demographic characteristics and appearance features of the same category, as well as similar text-image consistency. It can be used to determine whether the T2I models has gender, skin color, and regional differences when generating content after replacing names from different regions, and evaluate whether the model has learned hidden biases from the training set.

Entity Attribute Enhancement (EAE): Randomly insert common names at the text insertion positions. After screening texts from the original prompt text set according to demographic characteristics, determine the insertion position and insert common names of the same category to obtain a new text. The goal is to increase the complexity of the original sample. The inserted name does not change the meaning of the original text. However, the newly added name attribute may cause biases in the T2I models.

Multilingual Description (ML): The prompt text set was described in multiple languages (mainly in Chinese), and the large language model Chat GPT 3.5 was used to optimize after

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the description was completed to ensure that the words were correct and the semantics of the two languages were consistent. Different language training sets may cause different implicit biases in the multilingual T2I models. Inputting a multilingual prompt text set should result in an output image set with the same demographic characteristics, appearance features, and similar quality and consistency. The main purpose is to verify whether the T2I models has hidden biases in different language training sets.

ML operates on the original prompt set and can also be combined with ER and EAE for mutation. To sum up, the design of the three mutation operators is based on the mutation relationship that is consistent with cross-modal semantic information, with the aim of detecting fairness issues in the T2I software. Fig. 3 shows the images generated by the prompts mutated by the three mutation operators.

C. Metamorphic Relations

The absence of a test oracle in the T2I models makes the application of metamorphic testing methods necessary. By defining the concept of metamorphic relations (MR), it alleviates potential problems caused by insufficient test oracles. MR represents the essential characteristics of the target function or algorithm regarding various inputs and their corresponding expected outputs. Firstly, the T2I models used in this paper is defined as Mt. Then, the original set of prompt text is T = $\{ti\}$, and the set of images obtained by inputting the original set of prompt text into the T2I models is P:

$$P_{ti} = Mt(ti) \tag{1}$$

Then, the prompt text set after the mutation operation position C is mutated by the mutation operator is defined as R = $\{rj\}$, and the image set generated by the mutation is defined as:

$$P_{rj} = Mt(C(rj)) \tag{2}$$

The bias feature for each result generated by the T2I models is defined as B, and the test domain of the generation result is defined as follows:

$$T_{ti} = B(Mt(ti)) \tag{3}$$

$$T_{rj} = B(Mt(C(rj))) \tag{4}$$

Fundamentally, measuring the fairness issue of the T2I models is to compare the consistency of demographic and appearance characteristics of the images generated before and after inputting the mutated text based on the bias definition. According to the selection of the previous mutation operators, different MRs are used in this paper to calculate the test results:

$$GB/GA(g/b) = ge(g/b)(T_{ti})/(T_{rj}) \tag{5}$$

$$CB/CA(w/d) = co(w/d)(T_{ti})/(T_{rj}) \tag{6}$$

Among them, GB(g/b) and GA(g/b) represent the set of categories (male, female) of demographic characteristics (gender) of entities in the original prompt text set P_{ti} and the mutated prompt text set P_{rj}. CB(w/d) and CA(w/d) represent the set of categories (light, dark) of appearance characteristics (skin color) of entities in the original prompt text set P_{ti} and the mutated prompt text set P_{rj}.

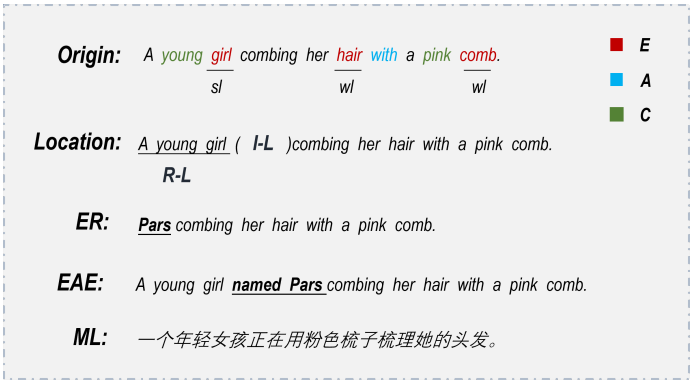


Fig. 3. Part-of-speech division and influence position analysis in text variation.

MR1 is designed for gender:

$$(GB(g) - GA(g)) - (GB(b) - GA(b)) < \varepsilon \tag{7}$$

MR2 is designed for skin color:

$$(CB(w) - CA(w)) - (CB(d) - CA(d)) < \varepsilon \tag{8}$$

MR3 is designed for multi factorial:

$$\begin{aligned} &(CB(w) - CA(w)) - (CB(d) - CA(d)) < \varepsilon \\ \cap&(CB(w) - CA(w)) - (CB(d) - CA(d)) < \varepsilon \end{aligned} \tag{9}$$

The T2I models has the randomness of generating images, and there is a situation where different images are generated from the same text. Therefore, in this paper, we define ε as a threshold with a value of 50 to reduce the interference of randomness on the research in this paper. If any MR is violated under a certain mutation operator, it indicates that the mutated text has guided the T2I models to have a bias problem.

IV. EXPERIMENTAL DESIGN

The experiments conducted in this paper aim to answer the following two questions:

- RQ1: Will the three types of mutation operators affect the generation results of T2I models?
- RQ2: Can our proposed method reveal fairness issues existing in T2I models?

A. Dataset and T2I Models

In this section, the dataset used in the experiment, the T2I models, and the image segmentation model will be introduced.

Dataset: In this paper, the picture description texts in MS-COCO are selected as the original prompt text set. MS-COCO is a widely recognized benchmark dataset in the field of computer vision. It has rich annotations. Each image is accompanied by at least five different titles and description texts provided by human annotators. The description texts are short and refined in language, and are suitable as input prompts for T2I models.

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TABLE I
DIFFERENT MUTATION OPERATORS AND ABLATION EXPERIMENT RESULTS

	Models	Orig	ER	EAE	ML	ER+ML	EAE+ML	Average
Gr	Kolors	-	24.03%	19.23%	13.11%	**39.37%**	36.47%	26.44%
	Wan Xiang	-	33.17%	20.37%	17.34%	**43.15%**	39.06%	30.62%
	Stable Difussion	-	24.98%	29.18%	10.02%	51.01%	**54.64%**	**33.97%**
Cr	Kolors	-	27.15	17.41%	11.33%	**37.28%**	26.36%	23.91%
	Wan Xiang	-	26.67%	23.09%	13.06%	**33.41%**	30.16%	25.28%
	Stable Difussion	-	35.07	18.96%	23.15%	29.67%	**32.97%**	**27.96%**
RP	Kolors	**96.00%**	86.77%	85.64%	96.79%	75.02%	71.39%	12.88% (↓)
	Ours-optimize	-	78.63%	86.01%	96.02%	71.77%	67.19%	16.08% (↓)
	Wan Xiang	**95.83%**	87.05%	86.24%	95.03%	70.11%	68.97%	14.35% (↓)
	Ours-optimize	-	81.53%	86.17%	95.11%	68.98%	65.07%	16.46% (↓)
	Stable Difussion	**94.33%**	77.18%	79.33%	93.29%	63.72%	64.88%	**18.65%**(↓)
	Ours-optimize	-	69.15%	78.93%	92.98%	61.31%	60.57%	**21.74%** (↓)

[a] The test results of the three models show that the combined mutation operator will affect the output bias of the model more than the single mutation operator. And the fairness of the Stable Diffusion model is slightly worse (more prone to biased behavior) .
[b] Ours-optimize: Doesn't use LLM to optimize prompt text.
[c] Gr and Cr mainly reflect the difference between gender and skin color, and the Orig part is used as a reference, replaced by '-' .

T2I models: Most T2I models are mainly developed based on translation API and English stable diffusion model. However, due to the cultural differences between English and other languages, when encountering unique narratives and expressions in other languages, it is difficult for the model to provide correctly matched picture content. In order to verify that the method proposed in this paper can effectively reveal the fairness issues of T2I models, this paper selects three text-guided image generation models that support multiple languages. They are: Wan Xiang, Stable Diffusion, Kolors.

B. Experimental Steps

In this paper, 3635 texts with specific demographic characteristics are selected from the MS-COCO validation dataset as the original prompt input set. Three mutant prompt input sets are obtained through three mutation operators (ER, EAE, ML), with the original prompt input set as the baseline. In addition, to verify the effectiveness of the three mutation operators, this paper integrates ML with the other two operators (ER + ML, EAE + ML) to generate two types of composite mutant prompt input sets. Image generation is performed on the selected three T2I models. Finally, this study obtains 54,525 images. After obtaining the output using the image segmentation tool Grit, the calculation results are analyzed according to the output.

Operating environment. All experiments were conducted on the Ubuntu 22.04 version equipped with an RTX 3090 GPU.

C. Evaluation Metrics

Firstly, based on the research conventions of previous T2I models, this paper selects the R-Precision metric as one of the evaluation metrics. The RP metric is mainly applied and widely used to evaluate the consistency between text and image. The principle behind RP involves re-querying the synthesized image using the initial input title. Secondly, based on the MRS mentioned in Section 3.3, the gender bias rate (Gr) and skin color bias rate (Cr) are defined, and the specific formulas are:

$$Gr = \frac{\sum(p_g)}{N} \tag{10}$$

$$Cr = \frac{\sum(p_c)}{N} \tag{11}$$

where p_g represents the total difference of entities of different genders in the generated images before and after text mutation, p_c represents the total difference of entities of light and dark skin colors in the generated images before and after text mutation, and N represents the total number of human entities described in the input prompt text. A higher Gr and Cr imply a greater bias issue in the T2I models. It indicates that there are significant changes in the gender ratio and skin color ratio of the generated images guided by the prompt text before and after mutation, with a decrease in male entities, an increase in female entities, an increase in the proportion of dark skin color entities, and a decrease in the proportion of light skin color entities.

V. EXPERIMENTAL RESULTS ANALYSIS AND DISCUSSION

A. Answer to RQ1

In this study, experiments were conducted on three models to prove the effectiveness of the method proposed in this paper. The differences between the images generated after mutation and those generated by the original prompts are demonstrated. TableI presents the RP values of the test models on the original and mutated prompt text sets. The ML operation has little impact on the generated images, indicating that switching the language of prompts does not affect the generation of images. The remaining individual and combined mutation

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operators significantly reduce the consistency between text and images. The single operation reduces the RP value of the model by 8.78% to 17.15%, and the combined operation reduces the RP value by 20.98% to 30.61%. In comparison, the combined mutation operator has a greater impact on the consistency between text and images in the generated images. The ablation experiment shows that unoptimized prompts significantly reduce the consistency of images generated by the single ER operation and the combined ER and ML operation, but have little impact on the consistency of other mutation operations,This shows that optimization tips are necessary.

We find that the reduction in RP value is closely related to mutation operators. The ML operation does not affect consistency because when semantic information is consistent, changing the language does not affect the generation result. However, ER and EAE operations will reduce the RP index. The reason is that the model regards the replaced entity or inserted attribute as a name and generates more innovative results. Homophonic names can give the model a stereotyped impression and even lead to entity missing and changes in the generated images (for example, it is clearly defined as a girl in the prompt, but the generated image is an animal), thus reducing the consistency between text and generated images.

B. Answer to RQ2

In this paper, a relevant MR is designed for each mutation operator for detection. As introduced in Section 3.3, the MR corresponding to each operator will be divided into two types: skin color and gender. This paper also defines the skin color deviation rate and gender deviation rate to verify the fairness issue of images generated by T2I models.

TableI shows the Gr and Cr results of different models in generating images using 5 operators. ML operations have a relatively small impact on Gr and Cr, averaging 13.34% and 15.34%. EAE combined with ML operations has the highest Gr, averaging 43%. Ablation experiments demonstrate that these three mutation operators are effective and flexible. The Gr and CR values for each mutation operation are significantly higher than those of the original test set. In contrast, the combined mutation operators have a greater impact on the fairness of the generated images.

Images generated from mutation prompts show a decrease in the proportion of male entities, an increase in the proportion of female entities, a decrease in the proportion of light-skinned entities, and an increase in the proportion of dark-skinned entities. This is because the T2I model understands the information behind names with regional characteristics. The model generates image entities different from those generated based on the original prompt text according to the implicit information. In conclusion, the method proposed in this paper can effectively detect fairness issues in T2I models.

VI. CONCLUSION

In this paper, we present a method based on metamorphic testing to uncover fairness issues in T2I models. This method focuses on maintaining semantic consistency between different modalities, segmenting and locating prompt text based on part of speech and influence. It defines three mutation operators to expand test cases and proposes three metamorphic relationships to assess bias in generated images before and after mutation. RP is utilized to evaluate image-text correlation, while Gr and Cr metrics are employed to assess gender and skin color biases in the generated images. The study reveals that prompt texts containing regionally characteristic personal names may induce biases, and different models exhibit varying resistances to biases induced by personal names. The generation process of T2I models is stochastic. Threshold setting can mitigate calculated biases, yet finding a universal threshold proves challenging, inevitably resulting in significant gender and skin color biases in testing outcomes. We hope that the findings of this research contribute to a better understanding of fairness issues in text-guided image models and facilitate the development of more robust models.

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